# SOCIOL 723: Social Statistics II

## Week 3, Maximum Likelihood: Theory

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# Today

Why We Need More Than Least Squares

The Likelihood Function

Finding the Maximum

Information and the Variance of the MLE

Properties of the MLE

Likelihood-Based Tests

★ = essential, you must understand this ★ = for understanding, not examined

# Why Likelihood

# What Least Squares Bought Us, and What It Cost ★

Two weeks, one idea: pick the line that makes the squared misses as small as possible. Look at how much that bought:

- A slope with a clear meaning (the best linear approximation to the CEF), computable with sums alone.
- Honest uncertainty for it, robust to heteroskedasticity and clustering, without ever assuming what the errors look like.

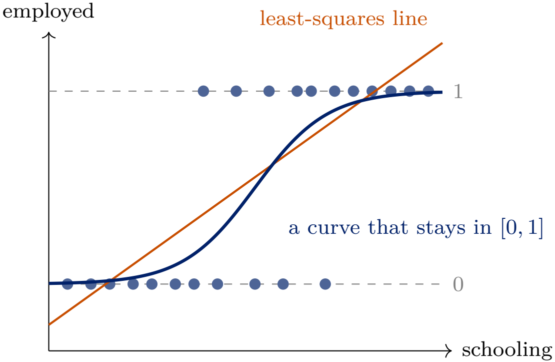
It asked almost nothing in return: never *how* $Y$ was generated, only *which line fits best*.

**That is also its limit**

Least squares cannot ask how were these outcomes produced? For earnings you can afford not to ask. For much of sociology's data, you cannot.

# Where the Line Runs Out ★

Regress "employed" (0/1) on years of schooling, by least squares:




- The line predicts "probabilities" below 0 and above 1: the shape of the outcome carries information the line cannot use.
- Counts go negative the same way; durations get censored; categories have no order. The problem is not the slope: a line is the wrong *kind* of answer.

# The Question Underneath ★

Least squares asks which line comes closest to the points. Maximum likelihood asks something different:

"Which parameter values would have made the data I observed most likely?"

To ask that, you have to say how $Y$ is produced: a coin for employment, a bell curve centred on a line for earnings. In return:

- One method for any outcome: binary, count, ordered, duration.
- Standard errors from the shape of the likelihood, not a new formula per model.
- One way to compare models, logit or Poisson alike: LR tests, AIC, BIC.
- Nothing lost: with bell-shaped errors it returns the least-squares line exactly.

Hence two weeks: this week the method, next week logit, probit, Poisson.

Stars: ★ examinable; ★ never examined.

# Two Stances, Both Legitimate

Two questions, two stances toward a regression; this course uses both.

- **Design-based** (Weeks 1–2, 5–11): $\beta$ is the best linear approximation to the CEF; assume little about how $Y$ was generated; let the *research design* carry the causal claim.
- **Model-based** (Weeks 3–4): write a probability model for $Y$ and estimate the parameter values that make the data most probable. Needed when the outcome is not a real number.

**Neither is "the right one"**

Say which stance a result rests on: a logit coefficient and a difference-in-differences estimate answer different kinds of question.

# The Likelihood

# Probability Versus Likelihood ★

Both start from the same object, the density (or mass function) $f(y \mid \theta)$. What differs is *what you hold fixed.*

**Probability**

$$f(y \mid \theta)$$

$\theta$ is *fixed and known*; $y$ varies.

"If the coin is fair, how likely am I to see 7 heads in 10 tosses?"

**Likelihood**

$$\mathcal{L}(\theta; y) \equiv f(y \mid \theta)$$

$y$ is *fixed* (it is your data); $\theta$ varies.

"Given that I saw 7 heads in 10 tosses, which values of $\theta$ make that outcome most probable?"

**The single most important sentence today**

The likelihood is *the same formula read backwards*: a function of $\theta$ with the data fixed. It is not a distribution over $\theta$: it does not integrate to one.

# A First Example: Marbles in a Bag ★

An opaque bag holds blue and white marbles; $\theta$ is the proportion blue. You draw 10 marbles *with replacement* and get 7 blue. Ten draws are ten *observations*, each a Bernoulli outcome (blue or not); the summary "7 of 10" is a binomial count. Build its probability step by step:

- **One draw**: blue with probability $\theta$, white with probability $1-\theta$ (replacement makes the draws identical and independent).
- **One particular sequence**, say B B B B B B B W W W: independence means multiply, $\theta \cdot \theta \cdots \theta \cdot (1-\theta)(1-\theta)(1-\theta) = \theta^7(1-\theta)^3$. Any other order with 7 blues and 3 whites has the *same* product.
- **All such sequences**: there are $\binom{10}{3} = \binom{10}{7} = \frac{10!}{3!\,7!} = 120$ ways to place the 3 whites (equivalently the 7 blues) among the 10 draws, so $\Pr(7 \text{ blue} \mid \theta) = 120\,\theta^7(1-\theta)^3$. ($\binom{n}{k}$, read "$n$ choose $k$," is the standard notation.)

The 120 does not involve $\theta$: it cannot change which candidate wins.

# Marbles: Reading It as a Likelihood

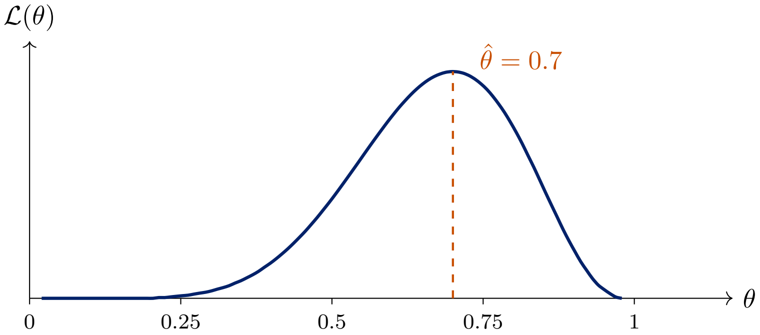
Now hold the data fixed, 7 blue in 10, and let $\theta$ vary:

$$\mathcal{L}(\theta) = 120\,\theta^7(1-\theta)^3.$$

- Try $\theta = 0.3$: $\mathcal{L} = 120(0.3)^7(0.7)^3 \approx 0.0090$.
- Try $\theta = 0.5$: $\mathcal{L} = 120(0.5)^{10} \approx 0.117$.
- Try $\theta = 0.7$: $\mathcal{L} = 120(0.7)^7(0.3)^3 \approx 0.267$.

Among these three, $\theta = 0.7$ makes the data we actually saw most probable. The MLE is the value that does best among *all* $\theta \in [0,1]$: the next frame plots the whole curve.

# The Likelihood, Plotted




- The height at any $\theta$ is the probability of *these* data if that $\theta$ were true.
- The peak is the MLE. The width of the peak will turn out to be our measure of uncertainty; that is the whole of Section 4.

# From One Observation to Many ★

With $n$ observations we need the *joint* density $f(y_1, \ldots, y_n \mid \theta)$. If the observations are independent given $\theta$, it factors:

$$\mathcal{L}(\theta; y_1, \ldots, y_n) = \prod_{i=1}^{n} f(y_i \mid \theta).$$

(Two observations first: independence means $f(y_1, y_2 \mid \theta) = f(y_1 \mid \theta)\, f(y_2 \mid \theta)$, the probability of both is the product; $n$ observations just repeat the step.)

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(Two observations first: independence means $f(y_1, y_2 \mid \theta) = f(y_1 \mid \theta)\, f(y_2 \mid \theta)$, the probability of both is the product; $n$ observations just repeat the step.) Products of many small numbers underflow and are painful to differentiate, so we take logs:

$$\boxed{\ell(\theta) \equiv \log \mathcal{L}(\theta) = \sum_{i=1}^{n} \log f(y_i \mid \theta)}$$

- log is strictly increasing, so it does not move the maximum.
- A product becomes a sum, so the derivative is a sum, which is why every MLE formula looks like a sample average.

# Worked Example 1: A Binomial Count ★

The marble game: $n$ Bernoulli draws, $Y_i \in \{0, 1\}$, $\Pr(Y_i = 1) = \theta$. The data that matter are the *count* of successes out of $n$; $\theta$ is the probability to estimate from it. One draw's mass function as a single formula:

$$f(y_i \mid \theta) = \theta^{y_i}(1-\theta)^{1-y_i}.$$

(At $y_i = 1$ it gives $\theta$; at $y_i = 0$, $1 - \theta$.)

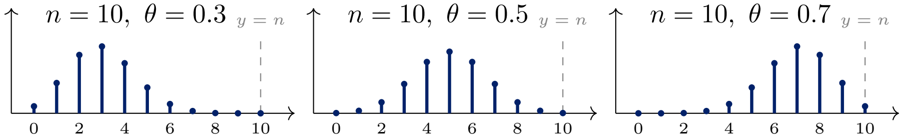
$$\mathcal{L}(\theta) = \prod_{i=1}^{n} \theta^{y_i}(1-\theta)^{1-y_i} = \theta^{\sum_i y_i}(1-\theta)^{n-\sum_i y_i},$$

$$\ell(\theta) = \Big(\textstyle\sum_i y_i\Big) \log\theta + \Big(n - \textstyle\sum_i y_i\Big) \log(1-\theta).$$

**Sequence or count?** This is the probability of the *particular* 0/1 sequence; the binomial multiplies it by $\binom{n}{\sum y_i}$ for the probability of the *count*. A constant factor, so the same MLE: for $\theta$, the count is all that matters (the data enter only through $\sum_i y_i$, a *sufficient statistic*). Counts are where we are headed.

# Worked Example 1: Reading the Binomial ★

$$\Pr(Y = y \mid n, \theta) = \binom{n}{y} \theta^y (1-\theta)^{n-y}, \qquad y = 0, 1, \ldots, n$$




- **The pieces**: $\theta^y$ for the $y$ successes, $(1-\theta)^{n-y}$ for the failures, $\binom{n}{y}$ for the orders they can come in.
- **The shape**: centred at $n\theta$, spread $n\theta(1-\theta)$, and a hard ceiling: no count above $n$ is possible.
- **Read down, not across**: the marble game fixes $y = 7$ and asks which panel puts the most mass on 7. The $\theta = 0.7$ panel wins: that is the likelihood.

# Counting Units or Counting Events? ★

Sociology's counts come in two kinds.

- **Units with an event, out of a known $n$**: 12 of 30 students protested. Each unit is a Bernoulli 0/1 (Worked Example 1); the total is binomial; it cannot exceed $n$.
- **Events one unit accumulates over exposure**: arrests of one person in a year; children; publications. There is no $n$ to count out of, and no ceiling.

**Rule of thumb**

First kind: binomial (in practice a logit on the 0/1, Week 4). Second kind: we need a distribution with no ceiling. Build it from the binomial, next.

# Worked Example 2: From the Binomial to the Poisson ★

Start from the binomial count of Worked Example 1:

$$\Pr(Y = y \mid n, p) = \binom{n}{y} p^y (1-p)^{n-y}.$$

Read it as *arrests of one person in a year*: slice the year into $n$ moments, each a Bernoulli trial with chance $p$ of an arrest. Then $y$ is the number of arrests: $p^y$ for the $y$ moments with one, $(1-p)^{n-y}$ for the moments without, $\binom{n}{y}$ for which moments they were.

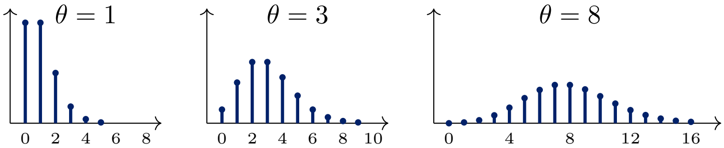
Now slice finer: 365 days, 8,760 hours, 525,600 minutes, … The slices $n \to \infty$ and the chance per slice $p \to 0$, but the expected arrests per year, $np = E[Y] = \theta$, cannot depend on how you slice. Fix $y$ and take the limit:

$$\binom{n}{y} p^y (1-p)^{n-y} \longrightarrow \frac{\theta^y e^{-\theta}}{y!}, \qquad y = 0, 1, 2, \ldots$$

The Poisson distribution: one parameter $\theta$, the expected count. Neither $n$ nor $p$ survives, so nothing caps the count.

# Worked Example 2: Reading the Poisson ★

$$\Pr(Y = y \mid \theta) = \frac{\theta^y \, e^{-\theta}}{y!}, \qquad y = 0, 1, 2, \ldots$$




- **The pieces**: $\theta^y$ grows with $y$, $y!$ shrinks it back (large counts are rare), $e^{-\theta}$ makes the probabilities sum to one.
- **The shapes**: piled at zero and right-skewed for small $\theta$; nearly a bell centred at $\theta$ for large $\theta$. No ceiling anywhere: the binomial's $n$ is gone.
- **The signature**: mean and variance are both $\theta$, so the spread grows with the mean: heteroskedasticity built in.

# Worked Example 2: The Poisson Log-Likelihood ★

Independent Poisson counts $Y_1, \ldots, Y_n$, common $\theta$; step by step:

$$\mathcal{L}(\theta) = \prod_{i=1}^{n} \frac{\theta^{y_i} e^{-\theta}}{y_i!} \qquad \text{independence: multiply}$$

$$\ell(\theta) = \sum_{i=1}^{n} \log\left(\frac{\theta^{y_i} e^{-\theta}}{y_i!}\right) \qquad \text{log turns the product into a sum}$$

$$= \sum_{i=1}^{n} \Big(y_i \log\theta - \theta - \log(y_i!)\Big) \qquad \log\theta^y = y\log\theta,\ \log e^{-\theta} = -\theta$$

$$= \log(\theta) \sum_i y_i \ - \ n\theta \ - \ \sum_i \log(y_i!) \qquad \text{collect: } n \text{ copies of } \theta$$

- The last term has no $\theta$: an *additive constant*, droppable for maximization but not when comparing log-likelihood *values* across models.
- Again the data enter only through $\sum_i y_i$: the total count is sufficient.

# Preview: Where the Regressors Go ★

So far every observation shares one $\theta$. Next week each observation gets its own, driven by covariates:

$$\theta_i = \exp(\beta_0 + \beta_1 X_i), \qquad \ell(\beta_0, \beta_1) = \sum_i \Big[ y_i(\beta_0 + \beta_1 X_i) - e^{\beta_0 + \beta_1 X_i} - \log(y_i!) \Big].$$

- The exponential keeps every $\theta_i$ positive, whatever the covariates do; this is the *log link* of Poisson regression.
- Same recipe: density, product, log, sum. Only the parameter changed from a number to a function of $X_i$.
- No closed form any more ($\beta$ sits inside and outside the exponential), so the maximum is found numerically: Section 3.

# Worked Example 3: The Normal Linear Model ★

The normal (Gaussian) density, a bell centred at $\mu$ with spread $\sigma$:

$$f(y \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y-\mu)^2}{2\sigma^2}\right).$$

Two assumptions from Week 1, now used together:

1. **Linear CEF**: $E[Y_i \mid X_i] = \beta_0 + \beta_1 X_i$.
2. **Normal errors**: $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$, independently.

Together they say $Y_i \mid X_i \sim \mathcal{N}(\beta_0 + \beta_1 X_i,\ \sigma^2)$: the bell's centre $\mu$ is the line. Put $\mu = b_0 + b_1 X_i$ into the density, and one observation's likelihood at candidate values $(b_0, b_1)$ is

$$f(Y_i \mid b_0, b_1) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(Y_i - b_0 - b_1 X_i)^2}{2\sigma^2}\right).$$

# Cashing the Hook: OLS Is an MLE ★

Product over independent observations, then log:

$$\ell(b_0, b_1) = \sum_{i=1}^{n} \log f(Y_i \mid b_0, b_1) = \sum_{i=1}^{n} \left[ -\tfrac{1}{2} \log(2\pi\sigma^2) - \frac{(Y_i - b_0 - b_1 X_i)^2}{2\sigma^2} \right]$$

$$= \underbrace{-\frac{n}{2} \log(2\pi\sigma^2)}_{\text{no } b_0, b_1} - \frac{1}{2\sigma^2} \underbrace{\sum_{i=1}^{n} (Y_i - b_0 - b_1 X_i)^2}_{\text{Week 1's } S(b_0, b_1)}.$$

- The candidates enter only through $S$, with a minus sign and a positive factor in front. Maximizing $\ell$ is exactly minimizing $S$: the MLE is the least-squares line, digit for digit.
- Least squares was never arbitrary: it is what "make the data as probable as possible" prescribes for normal errors around a line.

# What the Hook Does Not Say ★

- **OLS itself never needed normality.** Weeks 1–2 built it from moments alone, and its inference came from the CLT, not from any error distribution.
- **Normality is the price of admission** to the likelihood framework. For a continuous outcome it buys nothing new: OLS already gave the estimate, and Week 2 already gave its standard errors, with no distribution assumed.

**The point of the exercise**

Not "assume normality," but "one principle contains least squares as a special case and extends to everything least squares cannot do."

# Finding the MLE

# The Score Function ★

## Definition

The score is the derivative of the log-likelihood with respect to the parameter:

$$S(\theta) \equiv \frac{\partial \ell(\theta)}{\partial \theta} = \sum_{i=1}^{n} \frac{\partial \log f(y_i \mid \theta)}{\partial \theta}.$$

- The MLE $\hat{\theta}$ solves $S(\hat{\theta}) = 0$, the first-order condition, exactly as the normal equations were the FOC for least squares.
- Second-order condition: $\partial^2 \ell / \partial \theta^2 < 0$ at $\hat{\theta}$, so that we have a maximum rather than a minimum.
- A population fact: $E[S(\theta_0)] = 0$ (derived in Section 4). **Mind the direction**: here the true $\theta_0$ is fixed and the data vary, Week 2's sampling-distribution view, which is how any estimator is judged. Estimation runs the other way (data fixed, $\theta$ varies).

# Bernoulli: Deriving $\hat{\theta}$ ★

$$\ell(\theta) = \left(\sum_i y_i\right) \log\theta + \left(n - \sum_i y_i\right) \log(1-\theta)$$

$$S(\theta) = \frac{\sum_i y_i}{\theta} - \frac{n - \sum_i y_i}{1-\theta} \overset{!}{=} 0.$$

# Bernoulli: Deriving $\hat{\theta}$ ★

$$\ell(\theta) = \left(\sum_i y_i\right) \log\theta + \left(n - \sum_i y_i\right) \log(1-\theta)$$

$$S(\theta) = \frac{\sum_i y_i}{\theta} - \frac{n - \sum_i y_i}{1-\theta} \stackrel{!}{=} 0.$$

Multiply through by $\theta(1-\theta)$:

$$(1-\theta)\sum_i y_i - \theta\left(n - \sum_i y_i\right) = 0$$

$$\sum_i y_i - \theta\sum_i y_i - n\theta + \theta\sum_i y_i = 0$$

$$\boxed{\hat{\theta} = \frac{1}{n}\sum_i y_i = \bar{y}}$$

The MLE of a probability is the sample proportion, derived, not assumed.

# Poisson: Deriving $\hat{\theta}$ ★

Differentiate the log-likelihood and set the score to zero:

$$\ell(\theta) = \log(\theta) \sum_i y_i - n\theta - \sum_i \log(y_i!)$$

$$S(\theta) = \frac{\sum_i y_i}{\theta} - n$$

$\frac{d}{d\theta} \log\theta = \frac{1}{\theta}$; the constant's derivative is 0

$$S(\hat{\theta}) = 0 \implies \frac{\sum_i y_i}{\hat{\theta}} = n \implies \boxed{\hat{\theta} = \frac{\sum_i y_i}{n} = \bar{y}}$$

Second-order condition, to be sure it is a maximum:

$$\frac{\partial^2 \ell}{\partial \theta^2} = -\frac{\sum_i y_i}{\theta^2} < 0. \quad \checkmark$$

Both examples give $\bar{y}$, which is not a coincidence: for these distributions the mean *is* the parameter, and maximum likelihood finds it.

# Normal Linear Model: Deriving $\hat{\beta}$ and $\hat{\sigma}^2$ ★

Three parameters, so three first-order conditions. With $e_i = Y_i - b_0 - b_1 X_i$:

$$\frac{\partial \ell}{\partial b_0} = \frac{1}{\sigma^2}\sum_i e_i \overset{!}{=} 0, \qquad \frac{\partial \ell}{\partial b_1} = \frac{1}{\sigma^2}\sum_i X_i\, e_i \overset{!}{=} 0 \qquad \text{Week 1's normal equations, exactly}$$

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum_i e_i^2 \overset{!}{=} 0 \quad \Longrightarrow \quad \hat{\sigma}^2 = \frac{1}{n}\sum_i \hat{e}_i^2 \qquad \text{solve; } \hat{e}_i \text{ are the OLS residuals}$$

The first two conditions are Week 1's normal equations, so $\hat{\beta}_0, \hat{\beta}_1$ are the OLS estimates: the hook, cashed a second way.

### The MLE of $\sigma^2$ is biased

It divides by $n$, not $n-2$: $E[\hat{\sigma}^2] = \frac{n-2}{n}\sigma^2 < \sigma^2$. Week 1's $s^2$ was the bias-corrected version.

# When There Is No Closed Form

Most likelihoods you will meet, logit, probit, negative binomial, anything with covariates, have no algebraic solution to $S(\theta) = 0$. We search numerically.

## Newton–Raphson

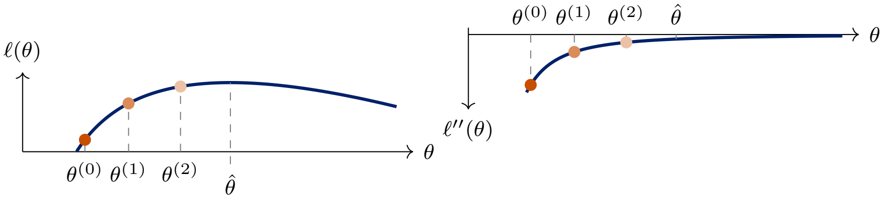
Start at $\theta^{(0)}$ and iterate

$$\theta^{(t+1)} = \theta^{(t)} - \left[ \left. \frac{\partial^2 \ell}{\partial \theta^2} \right|_{\theta^{(t)}} \right]^{-1} S(\theta^{(t)}),$$

a step of size score over curvature. The next frame draws it.

Variants: **BFGS** approximates the second derivative; **Fisher scoring** uses its expectation (what `glm()` does, as iteratively reweighted least squares).

# Finding the Peak: Newton–Raphson in a Picture




| | | | | |
|---|---|---|---|---|
| guess $\theta$ | 1.50 | 2.55 | 3.80 | 4.71 |
| slope $S$ | 2.33 | 0.96 | 0.32 | 0.06 |
| $-\ell''$ | 2.22 | 0.77 | 0.35 | 0.23 |
| step $S/(-\ell'')$ | 1.05 | 1.25 | 0.91 | 0.27 |

- Poisson, one observation $y = 5$: $\ell = 5\log\theta - \theta$; right panel $\ell'' = -5/\theta^2$. Step = slope $\div(-\ell'')$: gradient ascent, "learning rate" $1/(-\ell'')$ set by the data.
- The step is a *ratio*. Near the peak $-\ell''$ is smallest and the learning rate largest, yet the step shrinks: the slope goes to zero faster. Far out the slope is large, but so is $-\ell''$. At $\hat{\theta}$, $S = 0$ and $-\ell''(\hat{\theta})$ becomes the standard error.

# Practical Matters in R

- `optim()` minimizes. MLE is a maximization problem, so you pass it the *negative* log-likelihood.
- Supply `par`, the starting values. Poor starting values are the most common cause of failed convergence.
- Always check the convergence code (`$convergence == 0`), a returned number is not the same as a converged number.
- Ask for `hessian = TRUE`. The *Hessian* is the matrix of second derivatives of the function you optimized, computed numerically at the solution; with one parameter it is the single number $-\ell''(\hat{\theta})$ (sign flipped because you minimized $-\ell$), i.e. the observed information: your standard error.
- Try several starting values. If they land in different places, your likelihood is multimodal or your model is not identified.

All of this is Thursday's lab.

# Identification

**Definition**

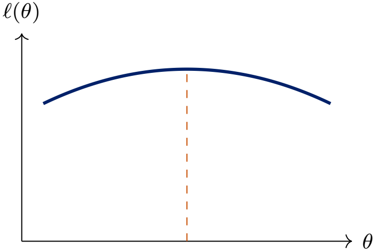
$\theta$ is identified if distinct parameter values imply distinct distributions of the data: $\theta^* \neq \theta \Rightarrow f(\cdot \mid \theta^*) \neq f(\cdot \mid \theta)$.

Non-identification means the likelihood is *flat* along some direction: the data cannot distinguish among a set of parameter values.

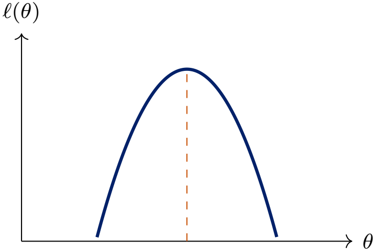
- **Classic example**: $Y_i \sim \mathcal{N}(\theta_1 - \theta_2, 1)$. Only the *difference* is identified. Optimizers started at different points will return different $(\hat{\theta}_1, \hat{\theta}_2)$ lying along a ridge, all with the same log-likelihood.
- This is the likelihood analogue of perfect collinearity in OLS, and it has the same symptom: a singular Hessian, so no standard errors.
- Diagnosis: multiple start values, and inspecting the Hessian's rank.

# Information

# Curvature Is Information ★




flat: little information




peaked: much information

Both log-likelihoods peak at the same $\hat{\theta}$. But on the left, many nearby values of $\theta$ fit almost as well; we are uncertain. On the right, moving away from $\hat{\theta}$ costs a lot of fit; we are confident.

The second derivative measures exactly this: the more negative, the more information.

# Fisher Information ★

**Definition (one parameter, at the true value)**

$$\mathcal{I}(\theta_0) \equiv -E[\ell''(\theta_0)] = E\left[S(\theta_0)^2\right] = \mathrm{Var}\,(S(\theta_0)).$$

Same direction as the Score frame: $\theta_0$ fixed, data random; $E$ and Var average over $y \sim f(y \mid \theta_0)$. Minus sign because $\ell'' < 0$ at a peak. $-E[\ell''] = E[S^2]$ is the information equality (orange frames; needs the model to be right); $E[S^2] = \mathrm{Var}(S)$ is only $E[S(\theta_0)] = 0$.

- *Additive over independent observations*: $\mathcal{I}_n = n\,\mathcal{I}_1$, the source of the familiar $1/n$.
- **From data**: plug $\hat{\theta}$ into your sample's own $-\ell''(\theta)$. This is the *observed information*, what software reports.

# Information Equality: Why Care ★

Four frames, one identity; its payoff is the variance of $\hat{\theta}$. At the peak the score is zero; expand it around the true $\theta_0$ (one Newton step, slope over curvature):

$$0 = S(\hat{\theta}) \approx S(\theta_0) + \ell''(\theta_0)\,(\hat{\theta} - \theta_0) \quad \Longrightarrow \quad \hat{\theta} - \theta_0 \approx \frac{S(\theta_0)}{-\ell''(\theta_0)}.$$

The denominator is a sum over observations, so by the LLN replace it by its expectation (Week 2's move): $\hat{\theta} - \theta_0 \approx c\,S(\theta_0)$ with $c = 1/(-E[\ell''(\theta_0)])$ fixed. Then $\mathrm{Var}(cS) = c^2\,\mathrm{Var}(S)$:

$$\mathrm{Var}(\hat{\theta}) \approx \frac{\mathrm{Var}\left(S(\theta_0)\right)}{\left(-E[\ell''(\theta_0)]\right)^2},$$ $\approx$: first-order expansion; exact in the $\sqrt{n}$ limit (Properties).

Two different quantities. The information equality, $\mathrm{Var}(S(\theta_0)) = -E[\ell''(\theta_0)]$, says both equal $\mathcal{I}(\theta_0)$, so the ratio collapses: $\mathrm{Var}(\hat{\theta}) \approx \mathcal{I}/\mathcal{I}^2 = 1/\mathcal{I}(\theta_0)$.

- One number, the second derivative at the peak, is the whole standard error: software inverts the Hessian at $\hat{\theta}$ and stops. Without the equality you would need both pieces separately.

# Information Equality: Goal and Plan ★

**What we want to show.** One observation $y \sim f(y \mid \theta)$ with its own score $s(y;\theta) \equiv \partial \log f(y \mid \theta)/\partial\theta = (\partial f/\partial\theta)/f$, at that same true $\theta$, random only through $y$, which $E$ averages over; the full score is $S = \sum_i s(y_i;\theta)$:

$$-E\left[\frac{\partial^2 \log f}{\partial\theta^2}\right] \;=\; E\left[s^2\right] \;=\; \mathrm{Var}(s).$$

**The only tool.** The density integrates to one at *every* $\theta$, so all its $\theta$-derivatives vanish:

$$\int f(y \mid \theta)\, dy = 1 \;\text{ for all } \theta \qquad \Longrightarrow \qquad \frac{d}{d\theta}\int f(y \mid \theta)\, dy = 0.$$

**Plan.**

1. Differentiate once $\Rightarrow$ $E[s] = 0$.
2. Differentiate twice $\Rightarrow$ $E\left[(\partial^2 f/\partial\theta^2)/f\right] = 0$.
3. Quotient rule on $s$ $\Rightarrow$ $\partial^2 \log f/\partial\theta^2 = (\partial^2 f/\partial\theta^2)/f - s^2$.
4. Take expectations of step 3; steps 1–2 leave only $s^2$.

# Information Equality: Steps 1–2 ★

**Step 1: differentiate once.**

$$0 = \frac{d}{d\theta} \int f(y \mid \theta)\, dy$$ derivative of the constant 1

$$= \int \frac{\partial f(y \mid \theta)}{\partial \theta}\, dy$$ move the derivative inside the integral

$$= \int s(y;\theta)\, f(y \mid \theta)\, dy$$ $s = (\partial f/\partial\theta)/f$, so $\partial f/\partial\theta = s\, f$; $s$ depends on $y$ too

$$= E\big[s(y;\theta)\big]$$ $\int g(y)\, f(y \mid \theta)\, dy = E[g(y)]$ for $y \sim f(\cdot \mid \theta)$, this same $\theta$

**Step 2: differentiate twice.** Line two above is 0 at every $\theta$; differentiate it again:

$$0 = \int \frac{\partial^2 f(y \mid \theta)}{\partial \theta^2}\, dy$$ derivative of a zero is zero

$$= \int \frac{\partial^2 f/\partial\theta^2}{f}\, f\, dy = E\left[\frac{\partial^2 f/\partial\theta^2}{f}\right]$$ multiply and divide by $f$: an expectation

# Information Equality: Step 3 ★

**Step 3: differentiate the score.** $s = (\partial f/\partial\theta)/f$ is a ratio, so use the quotient rule $(u/v)' = (u'v - uv')/v^2$ with $u = \partial f/\partial\theta$ and $v = f$:

$$\begin{aligned}
\frac{\partial^2 \log f}{\partial\theta^2} &= \frac{\partial s}{\partial\theta} && s \text{ is the first derivative of } \log f \\
&= \frac{\dfrac{\partial^2 f}{\partial\theta^2} f - \dfrac{\partial f}{\partial\theta}\dfrac{\partial f}{\partial\theta}}{f^2} && u' = \partial^2 f/\partial\theta^2,\ v' = \partial f/\partial\theta \\
&= \frac{\partial^2 f/\partial\theta^2}{f} - \left(\frac{\partial f/\partial\theta}{f}\right)^2 && \text{split the fraction; cancel one } f \text{ in each piece} \\
&= \frac{\partial^2 f/\partial\theta^2}{f} - s^2 && \text{the bracket is } s \text{ by definition}
\end{aligned}$$

An identity in $y$ and $\theta$, no expectation yet. The first term is the quantity step 2 showed has mean zero; the second is the squared score.

# Information Equality: Step 4 ★

**Step 4: take expectations of step 3.**

$$
\begin{aligned}
E\left[\frac{\partial^2 \log f}{\partial\theta^2}\right] &= E\left[\frac{\partial^2 f/\partial\theta^2}{f}\right] - E[s^2] && \text{expectation of a difference} \\
&= 0 - E[s^2] && \text{step 2 kills the first term} \\
&= -\operatorname{Var}(s) && \text{step 1: } E[s]=0, \text{ so } \operatorname{Var}(s) = E[s^2] - 0^2
\end{aligned}
$$

Multiply by $-1$: $-E\left[\partial^2 \log f/\partial\theta^2\right] = \operatorname{Var}(s)$. Done, for one observation.

- With $n$ independent observations, $\ell'' = \sum_i \partial^2 \log f(y_i \mid \theta)/\partial\theta^2$ and $S = \sum_i s(y_i;\theta)$: expectations add, and variances of independent terms add, so $-E[\ell''] = \operatorname{Var}(S)$ as well.
- Every expectation was taken under the *same* $f$ that integrates to one. If the data come from some other $g$, steps 1 and 2 fail, the two sides part ways, and the Hessian standard error is wrong.

# The Variance of the MLE ★

$$\boxed{\hat{V}(\hat{\theta}) = \left[ -\frac{\partial^2 \ell}{\partial \theta^2}\bigg|_{\hat{\theta}} \right]^{-1} = \frac{1}{\text{observed information}}}$$

Read it as a sentence: *the variance of the estimate is one over minus the second derivative of the log-likelihood at its peak*: the faster the slope changes there, the smaller the variance.

- With several parameters ★, the observed information is a matrix (the Hessian, sign flipped); invert it, and the standard errors are the square roots of its diagonal.
- This is why `optim(..., hessian = TRUE)` is not optional.

# Worked: Bernoulli Information

$$S(\theta) = \frac{\sum_i y_i}{\theta} - \frac{n - \sum_i y_i}{1-\theta},$$

$$\frac{\partial^2 \ell}{\partial \theta^2} = -\frac{\sum_i y_i}{\theta^2} - \frac{n - \sum_i y_i}{(1-\theta)^2}.$$

Taking expectations, using $E[\sum_i y_i] = n\theta$:

$$\mathcal{I}(\theta) = \frac{n\theta}{\theta^2} + \frac{n(1-\theta)}{(1-\theta)^2} = \frac{n}{\theta} + \frac{n}{1-\theta} = \frac{n}{\theta(1-\theta)}.$$

# Worked: Bernoulli Information

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$$\text{Var}(\hat{\theta}) = \mathcal{I}(\theta)^{-1} = \frac{\theta(1-\theta)}{n}. \quad \checkmark$$

Exactly the variance of a sample proportion you already know, now *derived* from the shape of the likelihood.

# Worked: Poisson Information

$$\frac{\partial^2 \ell}{\partial \theta^2} = -\frac{\sum_i y_i}{\theta^2} \quad \Longrightarrow \quad \mathcal{I}(\theta) = \frac{E[\sum_i y_i]}{\theta^2} = \frac{n\theta}{\theta^2} = \frac{n}{\theta},$$

so

$$\mathrm{Var}(\hat{\theta}) = \frac{\theta}{n}, \qquad \hat{V}(\hat{\theta}) = \frac{\bar{y}}{n}.$$

Check directly: $\hat{\theta} = \bar{y}$ and $\mathrm{Var}(\bar{Y}) = \mathrm{Var}(Y)/n = \theta/n$, because the Poisson mean equals its variance. ✓ Two routes, same answer; when a direct calculation exists, use it as a check.

# Properties

# The Four Properties

Under regularity conditions, as $n \to \infty$:

1. **Consistency** ★: $\hat{\theta} \xrightarrow{p} \theta_0$.
2. **Asymptotic normality** ★: $\hat{\theta} \overset{a}{\sim} \mathcal{N}(\theta_0, \mathcal{I}(\theta_0)^{-1})$.
3. **Asymptotic efficiency** ★: if the model is right, no regular estimator has a smaller asymptotic variance (the Cramér–Rao bound). Stronger than Gauss–Markov, which ranked only *linear* unbiased estimators.
4. **Invariance** ★: if $\hat{\theta}$ is the MLE of $\theta$, then $c(\hat{\theta})$ is the MLE of $c(\theta)$ for any continuous $c(\cdot)$. Wonderfully convenient: the MLE of the odds is the odds of the MLE.

The first three are asymptotic, none of them says anything about $n = 40$. Invariance, by contrast, holds exactly in any sample.

# Why Consistency and Normality Hold ★

The argument is a first-order Taylor expansion of the score around $\theta_0$:

$$0 = S(\hat{\theta}) \approx S(\theta_0) + \frac{\partial S(\theta_0)}{\partial \theta}(\hat{\theta} - \theta_0) \implies \sqrt{n}(\hat{\theta} - \theta_0) \approx \frac{\frac{1}{\sqrt{n}} S(\theta_0)}{-\frac{1}{n}\partial S(\theta_0)/\partial\theta}.$$

- The **denominator** is a sample average of second derivatives; by the LLN it converges to $\mathcal{I}_1(\theta_0)$.
- The **numerator** is a normalized sum of the i.i.d. mean-zero scores; by the CLT it is asymptotically $\mathcal{N}(0, \mathcal{I}_1(\theta_0))$.
- Combining: $\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}(0, \mathcal{I}_1(\theta_0)^{-1})$.

This is structurally identical to the OLS argument in Week 2: a sample average in the numerator, a sample average in the denominator, LLN below and CLT above.

# Maximum Likelihood Is Not Unbiased ★

- We already saw it: $\hat{\sigma}^2_{\text{MLE}} = \frac{1}{n}\sum_i \hat{e}_i^2$ has $E[\hat{\sigma}^2] = \frac{n-2}{n}\sigma^2$ in the bivariate model; $s^2$ divides by $n-2$ precisely to undo this.

**The general claim.** Maximum likelihood makes no promise about unbiasedness in a finite sample. Some MLEs happen to be unbiased (the Poisson $\hat{\theta} = \bar{y}$; $\hat{\beta}$ in the normal linear model, which is OLS); others are biased ($\hat{\sigma}^2$ above). What the method guarantees is that the bias shrinks to zero as $n$ grows: that is consistency, the first of the four properties.

**The trade we are making**

Give up exact unbiasedness in finite samples; get consistency, asymptotic efficiency, and a completely general recipe that works for any probability model you can write down.

# Testing

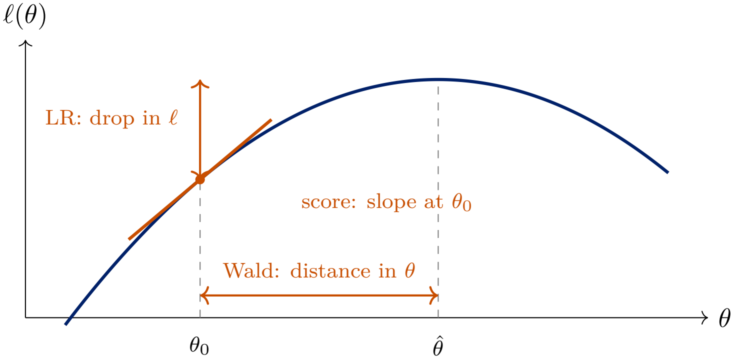
# Why Three Tests ★

The intuitive test is Week 2's: $z = (\hat{\theta} - \theta_0)/\text{se}(\hat{\theta})$, the **Wald** test. It is right whenever $\ell$ is a parabola, because then the curvature at $\hat{\theta}$ describes the whole curve. It only *knows* the curve at $\hat{\theta}$, and when the curve is not a parabola it can fail outright.

Twenty trials, zero successes: $\hat{\theta} = 0$ and $\text{se} = \sqrt{\hat{\theta}(1-\hat{\theta})/n} = 0$, so Wald cannot test $\theta_0 = 0.1$ at all. Yet $\ell(0.1) - \ell(0) = 20 \log 0.9 = -2.1$: imposing $\theta_0$ costs 2.1 units of fit, and $2 \times 2.1 = 4.2 > 3.84$ rejects. The likelihood knew; the Wald test could not see it. (A logit near separation fails the same way, Week 4.)

Hence two more distances. **Likelihood ratio**: the drop in $\ell$ between $\hat{\theta}$ and $\theta_0$. **Score**: the slope at $\theta_0$, no alternative fitted. All three agree when $\ell$ is a parabola, which large samples deliver. In your output: `summary(glm)` $z$ is Wald, `anova` on two `glm` fits is LR, overdispersion tests are score.

# Three Ways to Measure the Same Distance




All three ask: *is* $\theta_0$ *far from* $\hat{\theta}$*?* They are asymptotically equivalent but can differ in finite samples.

# The Likelihood Ratio Test ★

Compare the maximized log-likelihood with and without the restriction:

$$LR = -2\left[\ell(\hat{\theta}_{\text{restricted}}) - \ell(\hat{\theta}_{\text{unrestricted}})\right] \xrightarrow{d} \chi^2_q,$$

where $q$ is the number of restrictions.

- If the restriction is true, imposing it should cost little fit. The restricted $\ell$ is never higher, so $LR \geq 0$.
- **Requires both models.** Usually easy, and the best-behaved of the three in finite samples: prefer it when you can.
- R: `lmtest::lrtest(m0, m1)`, or `anova(m0, m1, test = "Chisq")` for `glm`.
- The models must be nested and fitted on the same observations. Silently dropping different rows to missingness invalidates the test.

$\chi^2_q$: the distribution of a sum of $q$ squared independent standard normals. With one restriction, $\chi^2_1$ is $\mathcal{N}(0,1)^2$, so the 5% critical value is $1.96^2 = 3.84$.

# The Wald Test ★

Measure the horizontal distance, scaled by the estimated standard error:

$$W = \frac{(\hat{\theta} - \theta_0)^2}{\hat{V}(\hat{\theta})} \xrightarrow{d} \chi^2_1, \qquad \text{or} \quad z = \frac{\hat{\theta} - \theta_0}{\mathrm{se}(\hat{\theta})}.$$

- This is the $t$-ratio you already read off every regression table.
- **Requires only the unrestricted model**, convenient when the restricted model is hard to fit.
- Multi-parameter version ★: $W = (\boldsymbol{R}\hat{\boldsymbol{\theta}} - \boldsymbol{q})'[\boldsymbol{R}\hat{V}(\hat{\boldsymbol{\theta}})\boldsymbol{R}']^{-1}(\boldsymbol{R}\hat{\boldsymbol{\theta}} - \boldsymbol{q})$, identical in form to Week 2.
- Caution: the Wald test is not invariant to how you parameterize the restriction. Testing $\theta = 1$ and $\log\theta = 0$ can give different answers. The LR test does not have this problem.

# The Score (Lagrange Multiplier) Test

Measure the slope of the log-likelihood *at the null*:

$$LM = \frac{S(\theta_0)^2}{\mathcal{I}(\theta_0)} \xrightarrow{d} \chi^2_q.$$

- Logic: if $\theta_0$ were the maximum, the slope there would be zero. A steep slope is evidence against it.
- **Requires only the restricted model**, useful when the unrestricted model is expensive to fit, e.g. testing for overdispersion without fitting the negative binomial.
- Many familiar specification tests are score tests in disguise, including Breusch–Pagan for heteroskedasticity.
- **Which to use?** Asymptotically the three are identical. In finite samples: LR if you can fit both models, Wald if only the unrestricted is available, score if only the restricted is.

# Model Comparison Beyond Nested Tests ★

LR, Wald and score all require *nested* models. For non-nested comparison:

$$AIC = -2\ell(\hat{\theta}) + 2k, \qquad BIC = -2\ell(\hat{\theta}) + k \log n.$$

- Both penalize fit by complexity; lower is better.
- BIC penalizes more heavily once $n > 7$, and increasingly so with $n$. BIC targets the "true" model if it is in the candidate set; AIC targets predictive accuracy.
- Neither is a hypothesis test; there is no $p$-value and no threshold.
- Comparable only across models fitted on *exactly the same* data with the same outcome scale.

# AIC and BIC in Research Practice ★

- **Typical uses**: non-nested specifications (income vs. log income; probit vs. logit), the number of classes in a latent class analysis, random-effects structures in multilevel models.
- **Reading ΔBIC** (Raftery 1995): 0–2 weak, 2–6 positive, 6–10 strong, $> 10$ very strong. Report the differences, not just the winner.
- **The trap**: both models must be fitted to the same observations. Listwise deletion changes the sample when specifications use different variables; fix the sample first.
- They referee between *theoretically motivated* candidates, not hundreds of searched specifications (Week 2's multiple-testing caution).

Week 12 returns to model selection with cross-validation.

# Where This Leaves Us

- A likelihood is the data's probability read as a function of the parameter. The MLE maximizes it; the score is the derivative set to zero.
- The curvature of the log-likelihood is information; its inverse is the variance.
- The MLE is consistent, asymptotically normal, and efficient, but not unbiased; every guarantee is asymptotic.
- LR, Wald, and score are three measures of one distance.

## Next

**Thursday**: code a likelihood and maximize it numerically in `R`.
**Week 4**: put covariates inside the likelihood, the generalized linear model, and the limited dependent variable models you will actually use.

# References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- Raftery, Adrian E. 1995. “Bayesian Model Selection in Social Research.” *Sociological Methodology* 25: 111–163. [the BIC reading sociologists use]